

Aditya Khandelwal

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[Email](#) · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

PROFESSIONAL SUMMARY

CS undergraduate with hands-on experience building LLM-powered systems, automated data pipelines, and backend APIs in Python. At Sentient Scripts, developed LLM-based information extraction modules and end-to-end intelligence pipelines with structured output parsing (Pydantic) and scheduled automation. Built multi-agent systems with LangGraph (PromptGraph) and LLM chains with LangChain (IceBreaker). Research contributor at Samsung R&D on time series forecasting with PyTorch; built concurrent vision systems at Infosys Springboard using Python multiprocessing.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL

AI / ML: LangGraph, LangChain, OpenAI API, Groq API, Gemini API, RAG Pipelines, Embeddings, Vector Search, FAISS, PyTorch, YOLOv8, OpenCV, Time Series Forecasting

Backend: FastAPI, REST APIs, asyncio, Multiprocessing, Pydantic, Microservices Architecture

Full-Stack: Next.js, React.js

Databases: PostgreSQL (Supabase), MySQL, Vector Databases (FAISS)

DevOps & Tools: Git, Docker, CI/CD Pipelines, Linux

Core Concepts: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

PROFESSIONAL EXPERIENCE

- **AI Engineer Intern** Aug 2025 – Mar 2026
Sentient Scripts, Discovery Lab SRMIST
 - Built multi-source data aggregation pipelines in Python that collected, normalized, and consolidated structured intelligence from external feeds for real-time monitoring and trend tracking.
 - Developed an LLM-based information extraction module that processed unstructured text to identify and structure domain-specific technical metadata, enabling automated correlation across large datasets.
 - Designed an end-to-end intelligence pipeline integrating scheduled data collection, LLM-driven correlation with structured output parsing (Pydantic), and automated report generation with trend visualizations.
- **Research Intern** Apr 2025 – Jan 2026
Samsung R&D Institute India Bangalore (Remote)
 - Conducted research on Time Series Forecasting using the TOTEM model with PyTorch across multiple benchmark datasets and forecasting horizons.
 - Studied tokenization strategies and transformer-based foundational model architectures to analyze forecasting accuracy and generalization performance.
- **Python Developer Intern** Nov 2025 – Jan 2026
Infosys Springboard (Remote)
 - Built a multi-zone real-time crowd counting system using YOLOv8 and OpenCV with optimized concurrent video stream processing via Python multiprocessing.
 - Designed a centralized monitoring dashboard with automated threshold-based alerting; profiled and optimized detection pipeline for reduced per-frame latency.

TECHNICAL PROJECTS

- **PromptGraph – Multi-Agent Prompt Architecture Generator** [\[GitHub\]](#)
Python, LangGraph, Groq API, Pydantic, FastAPI
 - Built a 6-agent LangGraph StateGraph pipeline that converts natural language project ideas into structured prompt architectures and production-ready Python scaffolding.

- Implemented a 3-layer retry mechanism (Tenacity exponential backoff, rate limiting, schema self-healing on invalid LLM output) that reduced token usage by $\sim 59\%$ via compact schema representation.
- Designed a Critic–Refiner quality loop evaluating prompts on 5 dimensions (clarity, robustness, measurability, compliance, creativity); 190-test suite covering unit, integration, and live API regression tests.
- Used Pydantic for structured output validation and checkpoint/resume across pipeline stages for fault-tolerant multi-stage LLM workflows.
- **IceBreaker – AI LinkedIn Profile Analyzer** [\[GitHub\]](#)
Python, LangChain, Gemini API
 - Built a LangChain-orchestrated AI agent that retrieves and analyzes LinkedIn profile data via external API, then generates structured summaries and conversation starters using Gemini API.
 - Implemented LLM chain with prompt templates, output parsers, and external API integration; applied rate-limit handling and mock-mode testing for reliable development iteration.
- **ZonePro – Full-Stack Productivity Platform** [\[GitHub\]](#)
Next.js, TypeScript, React, Supabase, PostgreSQL, Tailwind CSS
 - Built a full-stack productivity platform with YouTube course tracking, Pomodoro sessions, habit streaks, and real-time cloud sync across devices.
 - Integrated Supabase Auth and PostgreSQL for secure multi-device persistence; used YouTube Data API v3 for automated playlist extraction and progress tracking.

HACKATHONS & ACHIEVEMENTS

- **Code Zero Hackathon – 2nd Place** National Level
 – Secured 2nd place in a national-level hackathon by building a functional AI prototype under strict time constraints.

EDUCATION

- **SRM Institute of Science and Technology** **2023 – 2027**
 B.Tech in Computer Science and Engineering CGPA: 8.90
- **BVB Vidyashram**
 Senior Secondary (Class XII) 82.80%